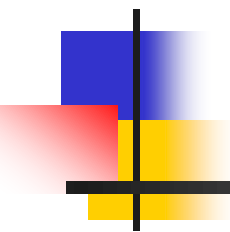


DATA STRUCTURES AND ALGORITHMS



Lecture Notes 1

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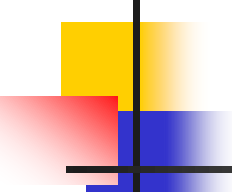
ROAD MAP

- **Data Types**
- Abstract Data Types
- Algorithms
- Data Structures
- Time & Space Complexity

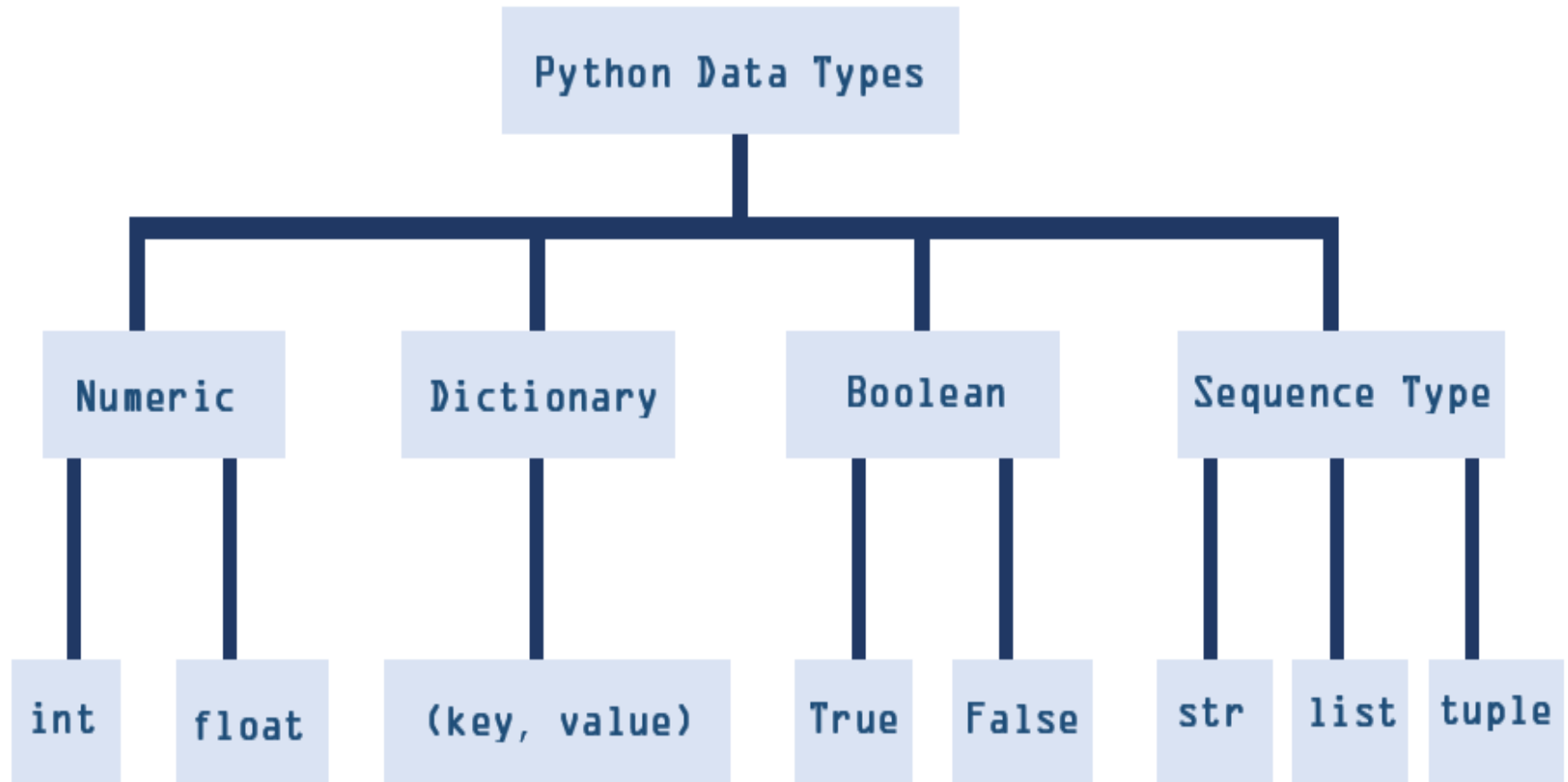


Data Types

- An attribute of data that informs the interpreter on how to classify the variable.
 - Python there are four categories of data types: numeric, sequential, booleans, and dictionary



Python Data Types





Abstract Data Types (ADT)

- A logical process on how we view data and allowed operations without the regard on how it will be implemented. (Mathematical abstraction)
- Think of an abstract data type as a process of the rules and operations of data.
 - There is no concrete implementation.



Algorithms

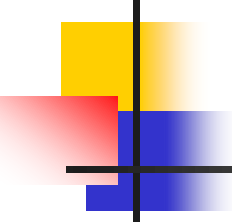
- A set of **step-by-step instructions** used to solve a problem.
- One example of an algorithm can be a recipe from a cookbook.
- When we follow **step-by-step instructions** from the cook book to create our meal.

Algorithms



- For an example "how to make omelet":
 - 1. Crack the eggs into a bowl.
 - 2. Melt the butter over medium-low heat, and keep the temperature low and slow when cooking the eggs.
 - 3. Add the eggs to the skillet and cook without stirring until the edges begin to set.
 - 4. Add the filling when the eggs begin to set. Cook for a few more seconds.
 - 5. Fold the omelet in half. Slide it onto a plate.





Data Structures

- A data structure is a format for accessing, storing, organizing, or structuring data.
- The implementation of an abstract data type can be referred to a data structure



Data Structures

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Data Structures and Abstract Data Types

- ADT gives us the blue print while a data structure tells us how to implement it.

Data Structures and Algorithms



- An algorithm processes data and that data is then stored into a data structure.

```
data = [0, 1, -1, 5] ← Data Structure
```

```
order = data.sort()  
        |  
    Algorithm
```



Time & Space Complexity

- Often, there is more than one way to solve the same problem with different programs. So **how would you be able to compare the performance of different algorithms**, is one program better than the other?



Time & Space Complexity

- There are two ways to determine which algorithm is more efficient:
 - The amount of space or memory an algorithm requires
 - The amount of time an algorithm requires to execute
- Overall time and space complexity can be impacted from several factors such as hardware, operating system.



Time & Space Complexity

- If we had to choose between either optimizing time or optimizing space.
- Overall, it depends on your needs, but in a production setting, **optimizing time is the main priority** because we can buy memory, but we can't buy time!
- This can lead to the best **trade off** of both increasing the space and lowering the time!